Points 6 Published

Details Questions

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Question 1.24 pts

Two alleles at one locus produce three distinct phenotypes. Two alleles of two genes lead to five distinct phenotypes. Two alleles of six genes lead to 13 distinct phenotypes. (These statements assume that the alleles at any one locus are codominant or incompletely dominant and that each gene makes an equal contribution to the phenotype.) Which of the following is a formula to express the relationship between number of genes and number of phenotypes? Let *n* equal the number of genes.

n
2n
2n + 1
3n
3n + 1

Question 1.2 pts

Each of the most extreme phenotypes for a trait determined by two alleles at one locus are found in a proportion of 1/4 in the F_2 generation. If two alleles of two genes determine the trait, each extreme phenotype will be present in the F_2 as 1/16 of the population.

In common wheat (*Triticum aestivum*) kernel color varies from red to white and the genes controlling the color act additively, that is, alleles for each gene are incompletely dominant and each gene contributes equally to the color. A true-breeding red variety is crossed to a true-breeding white variety, and 1/256 of the F_2 have red kernels and 1/256 have white kernels. How many genes control kernel color in this cross?

4

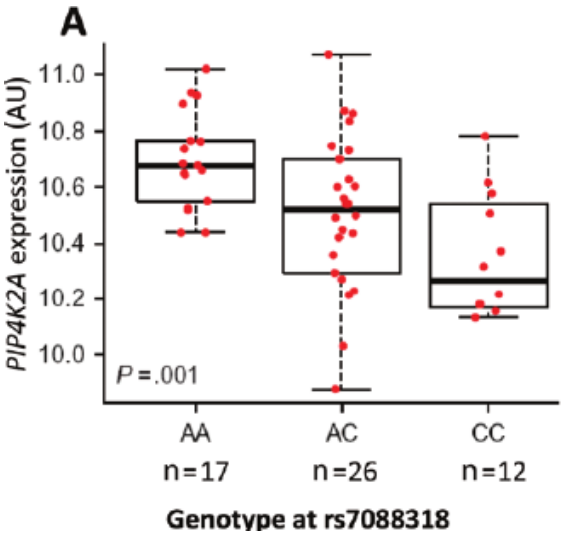
Question

Discussion (Chapter 22, Problem #16): To validate that the fw2.2 gene was the main causal gene for tomato size, researchers cloned the *S. pennellii* allele of the fw2.2 gene and introduced it as a transgene into *S. lycopersicum*. Why was the experiment done this way, rather than the other way around?

Question 0.5 pts

You performed a genome-wide study in lymphoblastoid cell lines derived from 55 individual donors to identify eQTLs. You examined 709509 SNPs. Using linear regression, you find an association of genotypes at rs7088318 with expression of the gene *PIP4K2A.* Shown below is a plot of the association along with the p-value of the association. AU = arbitrary units.

Is this a significant association?



Yes

No

Question 0.5 pts

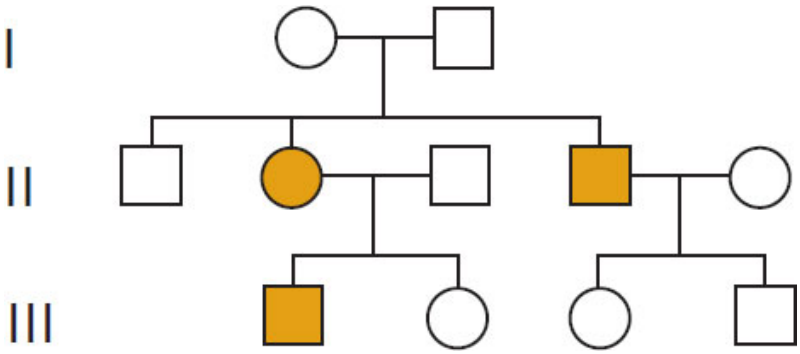
You sequence a short fragment of DNA using the ATAC-Seq method. Which of the following is true?

No transcription occurs in this region of the chromosome.

The chromatin is in a more open state than a part of the genome that was not sequenced with ATAC-Seq.

Question 1.06 pts

Assume that the disease illustrated with the pedigree below is due to the phenotypic expression of a rare recessive allele of an autosomal gene that is paternally imprinted. What would you predict is the genotype of individual I-1? Here, A is the normal dominant allele, and a is the recessive disease allele.



AA

Aa

aa

Question 0.5 pts

What is the predicted genotype of individual II-1?

AA

Aa

aa

Question 0.5 pts

What is the predicted genotype of individual III-2?

AA
Aa
aa

Question 0.5 pts

DNA regions that are transcriptionally silenced by hypercondensation of chromatin have often undergone a modification of their cytosine residues termed

hyperoxidation
methylation
dephosphorylation
phosphorylation
editing

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